Name_____ Date_____ Period_____

LESSON 7-1 · HOMEWORK PACKET · SHEET 1 OF 2

Integers and Opposites

Maryland Standard 6.NS.C.5 · Learning target – I can write an integer for a real situation and name its opposite.

REMEMBER

- **Integers**: … −3, −2, −1, 0, 1, 2, 3 …
- **Positive** = above 0 (gain, up). **Negative** = below 0 (loss, down).
- **Opposites** are the same distance from 0 on different sides: 6 and −6.

WORKED EXAMPLE

A diver swims 18 feet below sea level. Write an integer for her depth and its opposite.

Step 1 Below sea level is less than 0: −18.

Step 2 The opposite is 18 units from 0 on the other side: 18.

Answer **−18 feet; opposite 18**

PRACTICE

show your work · label units

1. Write an integer for each situation.

a. a loss of 12 yards _____

b. a deposit of $40 _____

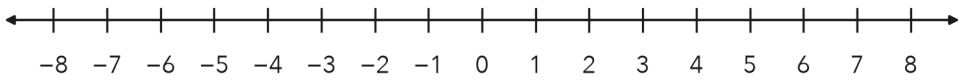
c. 9 floors below the street _____

d. 15 degrees above zero _____

e. a $25 withdrawal _____

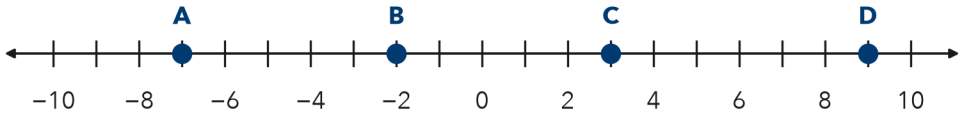
f. a gain of 7 pounds _____

2. Plot −6 and its opposite on the number line. Label each point with its value.



How far is each point from 0? _____ units

3. Write the integer for each point. Then write its opposite.



Point	A	B	C	D
Integer				
Opposite				

4. **Find the Error.** Keisha says the opposite of −15 is −15, because opposites keep the same number. Explain her mistake. Write the correct opposite.

_____

_____

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

AT HOME With someone at home, name a temperature on a weather report. Then name its opposite.

Name_____ Date_____ Period_____

LESSON 7-1 · HOMEWORK PACKET · SHEET 2 OF 2

Integers in Real Situations

Maryland Standard 6.NS.C.5 · Learning target – I can interpret integers in real situations and find the opposite of an opposite.

REMEMBER

- Positive: **above, gain, deposit**. Negative: **below, loss, withdrawal**.
- **0** is the start: sea level, a $0 balance.
- The **opposite of the opposite** is the number itself: −(−8) = 8.

WORKED EXAMPLE

Find the opposite of the opposite of 14.

Step 1 The opposite of 14 is −14.

Step 2 The opposite of −14 is 14.

Answer **14**

PRACTICE

show your work · label units

1. Write an integer for each situation.

a. a town 35 feet below sea level ____

b. earning $20 walking dogs ____

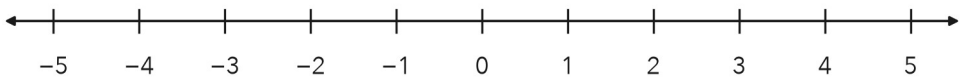
c. 4 degrees below zero ____

d. a loss of 8 yards ____

2. Find each value.

a. −(−9) = _____ **b.** −(15) = _____ **c.** −(−(−4)) = _____

3. Start at −3. Plot point **A** at −3, point **B** at the opposite of A, and point **C** at the opposite of B.



What do you notice about points A and C? _____

4. **Weather.** At noon it was 6 degrees above zero. At midnight it was 6 degrees below zero.

a. Write an integer for each temperature. _____

b. How are the two integers related? _____

5. **Reasoning.** Marcus says the opposite of a number is always negative. Is he right? Give an example.

_____

_____

TEST PRACTICE

choose the best answer

6. Which integer describes a loss of 18 points in a game?

A 18 **B** −18 **C** −(−18) **D** 0

7. Which equation is true?

A −(−11) = −11 **B** −(−11) = 11 **C** −(11) = 11 **D** −(0) = 1

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Tell someone at home a money story with a negative number. Ask what 0 means in it.